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AI Safety, Mechanistic Interpretability, Adversarial Robustness, White-Box AI Control

RESEARCH FOCUS

AI safety researcher working on mechanistic interpretability and adversarial robustness. Current focus is white-box monitoring for AI control: building activation-level monitors, then attacking them with adaptive attackers. Core methods are QK/OV circuit decomposition, direct logit attribution, activation patching, steering vectors, SAE feature attribution, and spectral dynamics. Work spans deception and scheming detection, safety-circuit analysis, red-teaming of mechanistic defenses, and AI-biosecurity at the representation layer. The approach throughout is understand, exploit, then defend.

RESEARCH EXPERIENCE

MARS V, Meridian AI Research Summer Jul 2026 – Present

Fellow, Lindley Lentati Track: White-Box Techniques for AI Control. Cambridge

- Trained linear compliance probes on residual-stream activations across four instruction-tuned models from three labs (Qwen2.5-7B, Llama-3.1-8B, Gemma-2-9B, abliterated Llama-3). Compliance is linearly decodable and tracks refusal itself rather than surface phrasing; length-matched accuracy is indistinguishable across models.
- Ran a red-team/blue-team co-evolution loop pitting adaptive attackers against those probes across three attack families: suffix-gradient, discrete search, and activation steering. The main result is negative. **Jailbreak success and probe evasion turned out mutually exclusive**: perturbations that evade the monitor also stop jailbreaking.
- Extending this into trained attack policies (Redwood open problem: stress-testing white-box monitors), gated behind a six-stage empirical de-risking protocol. Identified the system-prompt confound, whether probes read behaviour or instructions, as the primary failure mode, and designed the ablation that separates them.
- Multi-GPU SLURM cluster (8×B200), shared codebase with one collaborator.

Redarc Labs 2026 – Present

Co-founder and Research Lead. Delhi

- Founded and run a six-person AI safety research fellowship. Built Cohort 01 around one thesis: expressed confidence in LLMs is separable from internal knowledge, implemented partly through entropy neurons that modulate output entropy without changing token predictions.
- Mentoring three first-author workshop projects: attributing hedging to its cause (ignorance, safety training, sycophancy, sandbagging) from internal states; whether confidence regulation is shared across uncertainty types and survives post-training; whether white-box monitors for that channel can be evaded. Two papers under review at NeurIPS 2026 workshops.
- Run *Redarc Fellows*, an 8-week hands-on mechanistic interpretability cohort taught with PyTorch hooks and Hugging Face directly, no wrapper libraries.

Apart Research Nov 2025 – Present

Research Fellow. Mentor: Jason Hoelscher-Obermaier. Lead author, accepted at TAIS 2026, Oxford

- Measured geometric divergence of ViTs and ResNets under seven adversarial attack algorithms using Hutchinson's stochastic Hessian trace estimator. Found a five-order-of-magnitude discrepancy between input-space and parameter-space curvature, showing that standard robustness metrics misclassify internal model instability.

Artificial Intelligence Institute of South Carolina Jan 2025 – Apr 2026

Research Intern. Mentor: Dr. Amitava Das

- Co-author of AMBEDKAR (arXiv:2509.02133, cited by MIT Technology Review). Designed the fairness-aware speculative decoding algorithm, which reduces religion bias by 77% and caste bias by 23% at 6% latency overhead.

SELECTED PUBLICATIONS AND PROJECTS

Isolating Refusal and Compliance Heads: Unconditional Bipolar Defenses 2026

Safety circuits. Qwen2.5-7B. repo

- Compliance-score classifier reaches AUC = 0.943, Cohen's d = 2.636, with zero false positives across five benign categories: a working zero-shot white-box harm detector.
- Unconditional bipolar assertion (amplify refusal heads, zero-ablate compliance heads at L25-H4) reaches 33% ASR under nanoGCG, against 36 to 42% for conditional gating and 66% undefended. The precision of conditional gating creates an exploitable structural vulnerability under white-box optimisation.
- Found extreme intra-layer antagonism at Layer 25 between refusal and compliance heads. Includes Crescendo multi-turn analysis.

Pre-commitment Probing in Open-Weight Reasoning Models 2026

Mechanistic interpretability. OLMo-3-7B-Think. repo

- A linear probe on the residual stream predicts answer correctness before any reasoning token is emitted (AUC = 0.780, 5-fold CV). The model commits before it reasons, which makes CoT post-hoc rationalisation rather than deliberation.
- A functional frustration direction at Layer 15 transfers cross-lingually across six languages ($p = 0.0017$): a sub-linguistic computational state that is invisible in output but readable mechanistically, usable either as a monitor or as an attack surface.

Toxin Feature Hierarchy in ESM-2: Why Frozen Probes Resist ProteinMPNN Redesign 2026

GenBio 2026 poster, with Manan Wadhwa. repo

- BLAST detects 0% of 643 ProteinMPNN redesigns; a frozen ESM-2 linear probe detects 93.9% with no retraining. SAE analysis shows redesign amplifies toxin structural features rather than obscuring them.
- A four-tier attack taxonomy places the security boundary at gradient access: 6.1% evasion black-box against 100% white-box. Direct probe attribution pinpoints Layer 32 as the detection bottleneck ($r = 0.992$).

Context Poisoning in Tabular ICL: Geometric Hijacking and Inverse Scaling 2026

Mechanistic interpretability. Tabular foundation models. repo

- First logit-lens implementation for tabular ICL: a classification phase transition at Layer 8, representation hijacking as early as Layer 4, and total hijacking from 1.2% poisoned context. Larger models are more vulnerable, not less.
- QK/OV decomposition and DLA show that mechanistically-informed defenses (PCA subspace scrubbing, head ablation) fail because of manifold redundancy.

Loss Landscape Response to Adversarial Perturbation Is Architecture-Dependent 2026

TAIS 2026 workshop. Lead author, with Jason Hoelscher-Obermaier. repo

Spectral Regularization as a Safety-Critical Inductive Bias 2025

NeurIPS 2025 Reliable ML Workshop. Sole author. repo

- Fourier Gradient Regularization penalises high-frequency input-gradients during training, doubling adversarial robustness on ResNet-18 with CIFAR-10 at minimal clean-accuracy loss.

What You Pretrain On Matters: Synthetic Task Distributions Determine Tabular Foundation Model Quality 2026

FMSD at ICML 2026, accepted poster. With Bouadi, Bouarour, Kulkarni, Tanna, Sankarapu

- Introduced O'Prior, a compositional realism prior. Ran the interpretability and data-quality ablations confirming that mechanism diversity, realism composition, and shift-aware stress each contribute independently.

Physics-Informed Neural Operators with Multi-Method Explainability for Ocean Temperature Prediction 2026

AAAI 2026 Workshop on XAI4Science, accepted. Sole author

Known by Construction 2026

Model-state instrument calibration. Submitted. Equal co-contributor

OPEN-SOURCE TOOLING

- **neuron_autolabel**: auto-interpretability package for LLaMA-style models. Runs a hypothesis, test, revise loop per neuron, scored by detection AUC (logit gap) rather than simulation, with held-out splits and shuffled controls.
- **dead-salmon-fpr**: calibrated null-model test for interpretability claims. Naive significance testing flags up to 100% of findings on untrained models; a $K \geq 8$ to 12 null distribution corrects the false-positive rate.
- **mechanistic-tamper-verification**: two-tier integrity verifier (cryptographic plus mechanistic differencing) that detects tampered model updates by the concentration of weight change rather than its magnitude.

EDUCATION

Indian Institute of Technology Madras 2023 – 2027

BS in Data Science. Focus: AI safety, interpretability, fairness, governance

Maharaja Agrasen Institute of Technology, Delhi 2022 – 2026

B.Tech in Mechanical and Automation Engineering

TECHNICAL SKILLS

Languages: Python, CUDA, MATLAB, Bash, JavaScript

Frameworks: PyTorch, JAX, TransformerLens, NNsight, Hugging Face, DevInterp, LangChain, Docker, SLURM (multi-node, 8×B200)

Methods: QK/OV circuit decomposition, direct logit attribution, activation patching, steering vectors (CAA), logit lens, linear probing, SAE feature attribution, nanoGCG and suffix-gradient attacks, Hutchinson Hessian trace, SVD and spectral dynamics